# Lab 2-2: Answer Questions

**3.2** This exercise concerns TM $M_1$ whose description and state diagram appear in Example 3.9. In each of the parts, give the sequence of configurations that $M_1$ enters when started on the indicated input string.

[A]**a.** 11.
**b.** 1#1.
**c.** 1##1.
**d.** 10#11.
**e.** 10#10.

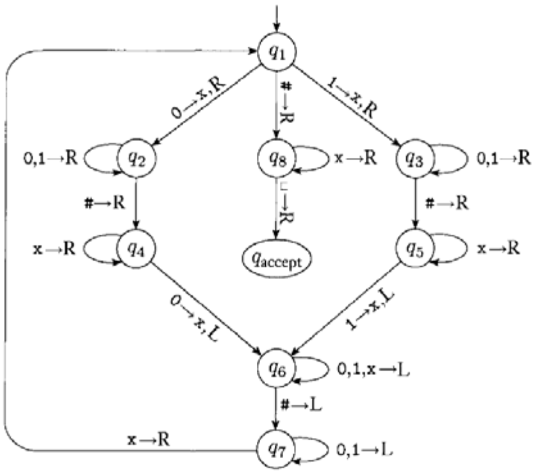


**FIGURE 3.10**
State diagram for Turing machine $M_1$

**3.8** Give implementation-level descriptions of Turing machines that decide the following languages over the alphabet $\{0,1\}$.

[A]**a.** $\{w \mid w$ contains an equal number of 0s and 1s$\}$
**b.** $\{w \mid w$ contains twice as many 0s as 1s$\}$
**c.** $\{w \mid w$ does not contain twice as many 0s as 1s$\}$

3.15 a,b,d,e.

**3.15** Show that the collection of decidable languages is closed under the operation of

[A]**a.** union.
**b.** concatenation.
**c.** star.
**d.** complementation.
**e.** intersection.

3.16 a,b,d.

**3.16** Show that the collection of Turing-recognizable languages is closed under the operation of

[A]**a.** union.
**b.** concatenation.
**c.** star.
**d.** intersection.